

Exercise 1:

Develop a dot net application with Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club fund. Generate pay slips for the employees with their gross and net salary.

Exercise 2:

Write a c# program that checks whether a given string is a palindrome or not. Ex MADAM is a palindrome

Exercise 3:

Write a class Book with three data members BookId, Pages and Price. It also contains the following member function: The get() method is used to input values The show() method is used to display values The set() method is used to set the values of data members using parameters The getPrice() method is used to return the value of price

Exercise 4:

Create a class building that has the public member floors,area, and occupants and a method areaperperson()respectively that display the area per person for building. In the main() method create two instance of building called house and office and display the area per person by division of area/occupants.

Exercise 5:

Write a class Result that contains roll no, name and marks of three subjects. The marks are stored in an array of integers. The class also contains the following member functions. • The input() method is used to input values • The show() method is used to display values • The total() returns the total marks a student • The avg() method returns the average marks of a student

Exercise 6:

Create a class rectangle. The class has attributes length and width each of which defaults to 1. It has member function that calculates the perimeter and the area of the rectangle. It has set() and get() functions for both length and width. The set() function should verify that length and width are each floating point numbers larger than 0.0 and less than 20.0.

Exercise 7:

Create a SavingsAccount class and use a static data member to contain the annualInterest for each of the savors. Each member of the class contains a private data member savingbalance indicating the amount the saver currently has deposited. Provide a monthlyInterest() member function that calculate the monthly interest by multiplying the balance by annualInterest by 12, this interest should be added to the savingbalance. Provide a static member function modifyInterest() that sets the static annualInterest to a new value. Write a driver program to test class SavingAccount and instantiate two objects saver1 & saver2 with balance of \$30000 and \$50000 respectively. Set annual interest 5% then calculate the monthly interest and show the new balance for each saver.

Exercise 8:

Create a new class SalariedEmployee that extends from class Employee and overrides the method earnings() of class Employee to calculate the fixed monthly salary of each employee. Create object of SalariedEmployee class in EmployeeTest class (i.e., main class) to set the record for new employee and print. Create a new method CalculateCommission in the CommissionEmployee class. This method calculates the commission based on the value of commissionRate data member.

Exercise 9:

Create three classes Food, Fruit and Apple in such a way that class Apple extends Fruit class and Fruit class extends class Food. Create a constructor in each class containing only a statement to print current class name.

Exercise 10:

Define a base class called Worker that has two derived classes: Employee and Manager. For workers, we only have simple attributes such as their names and working hours. For employees and managers, each has specific attributes unique to the class. Employees have attributes such as the name of the department that they are working with, and their salaries. Managers have attributes such as the number of managed employees and the number of projects that they're in charge. Also, define another class called Project that has attributes such as the name and the current progress (in %) of the project. (1) Write the corresponding constructors that initialize the attributes for classes. (2) Write the corresponding member functions to set and get the attributes for classes. (3) Finally, test the classes in a main program. In the test program, you should dynamically allocate some employees and managers, print their corresponding information.

Exercise 11:

A company pays its employees on a weekly basis. The employees are of four types: Salaried employees are paid a fixed weekly salary regardless of the number of hours worked, hourly employees are paid by the hour and receive overtime pay (i.e., 1.5 times their hourly salary rate) for all hours worked in excess of 40 hours, commission employees are paid a percentage of their sales and base-salaried commission employees receive a base salary plus a percentage of their sales. For the current pay period, the company has decided to reward salaried-commission employees by adding 10% to their base salaries. The company wants to write an application that performs its payroll calculations polymorphically.

Exercise 12:

Create an Abstract Class named Restaurant Create a function/method that will print the name of the restaurant when called. Create an abstract function/method named total price Create an abstract function/method named menu items Create an abstract function/method name location Create a Class called McDonalds that extends Restaurant Implement all abstract methods Add logic so that the total price method/function will give the total price of the meal including a 6% tax Add a method that returns a Boolean named hasPlayPlace, which returns true when this location has a play place Create a Constructor that will set the name of the Mcdonalds, location, and has PlayPlace

Exercise 13:

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Exercise 14:

Declare a Interface, RegisterForExams that contains single method register, implements the interface in two different classes (a) Student (b) Employee. Write down a Test Application that contains at least a function that takes Interface type Parameter

Exercise 15: Implement a Clock class that simulates time in hr:min:sec, derive a child class that overrides the display method and displays the time in both AM/PM and 24 hour format.

Exercise 16: Create an Abstract class Student that contains a method take exam, implement the method in the child classes PhdStudent and GradStudent in which PhdStudent takes exam by giving his final defense presentation while the graduate student gives a written paper.

Write a C# Sharp program to print Hello and your name in a separate line.

Write a C# Sharp program to print the sum of two numbers.

Write a C# Sharp program to swap two numbers

Write a C# Sharp program to print the output of multiplication of three numbers which will be entered by the user.

Write a C# Sharp program to print on screen the output of adding, subtracting, multiplying and dividing of two numbers which will be entered by the user.

Write a C# Sharp program that takes a number as input and print its multiplication table.

Write a C# Sharp program that takes four numbers as input to calculate and print the average.

Write a C# Sharp program that takes an age (for example 20) as input and prints something as "You look older than 20".

Write a C# Sharp program to accept two integers and check whether they are equal or not.

Write a C# Sharp program to check whether a given number is even or odd.

Write a C# Sharp program to check whether a given number is positive or negative.

Write a C# Sharp program to find whether a given year is a leap year or not.